(1)

$2 \times 32 = 64$ elements per read

$32^2 \times 64 = 65536$ total elements $\times 4$ bytes $=$

262144 bytes of dram traffic

(2)

$$\frac{32 \times 32}{8 \times 8} = 4 \text{ tiles}$$

$$\frac{32 \text{ wide}}{8 \text{ size}} = 4 \text{ steps}$$

$4 \times 4 \times 8 = 128$ tile loads

8×8 elements per tile $\times 4$ bytes $= 256$ bytes/tile

$128 \times 256 = 32768$ total bytes

③ $\frac{262144 \text{ naive traffic}}{32768 \text{ tiled traffic}} = \frac{8}{1}$ traffic ratio

$$\frac{N}{T} = \frac{32}{8} = \frac{4}{1}$$

Therefore the traffic ratio $= 2(N/T)$

④

$2 \times N^3 = 65536$ Flops

$\frac{65536}{32768} = 2$ Flops/byte

$\frac{10 \text{ TFlops}}{320 \text{ GB/s}} = 31.25$ Flop/Byte = ridge point

memory bound

$\frac{262144}{320 \times 10^9} = 8.192 \times 10^{-7}$ seconds

$\frac{32768}{320 \times 10^9} = 1.024 \times 10^{-7}$ seconds